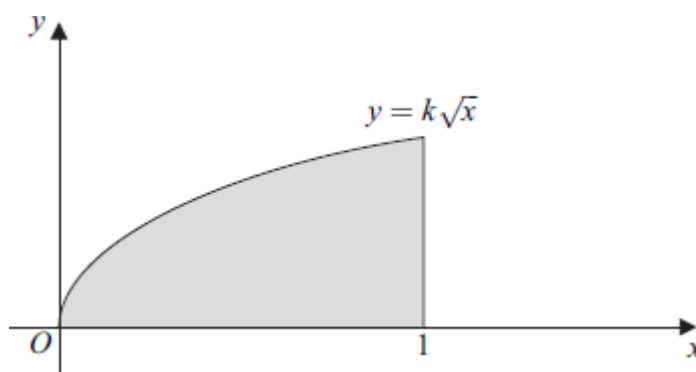


Q1.

A uniform lamina is bounded by the positive x -axis, the line $x = 1$ and the curve with equation $y = k\sqrt{x}$, where $k > 0$. The lamina is shown in the diagram.

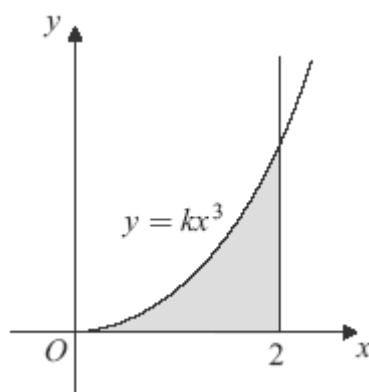


The area of the lamina is A square units.

- (a) Show that the x -coordinate of the centre of mass of the lamina is $\frac{2k}{5A}$. (3)
 - (b) Find, in terms of k and A , the y -coordinate of the centre of mass of the lamina. (3)
 - (c) Given that the centre of mass of the lamina lies on the line $y = x$, find the value of k . (2)
- (Total 8 marks)**

Q2.

A uniform lamina is bounded by the curve $y = kx^3$, the line $x = 2$ and the x -axis, as shown in the diagram.

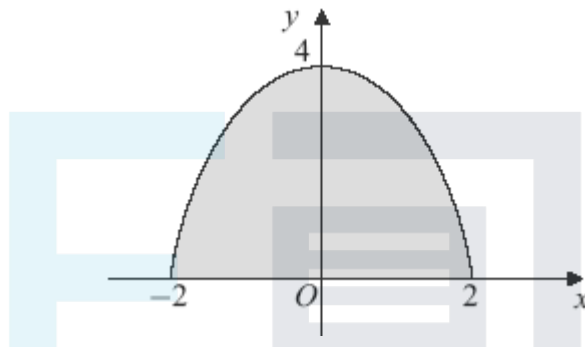


- (a) Find an expression for the area of the lamina in terms of k . (2)
- (b) Find the x -coordinate of the centre of mass of the lamina. (4)

- (c) The y -coordinate of the centre of mass of the lamina is 8.
- (i) Determine the value of k . (4)
- (ii) The lamina is freely suspended from the corner at the origin O . Find the acute angle between the straight edge at the point of suspension and the vertical. (3)
- (Total 13 marks)

Q3.

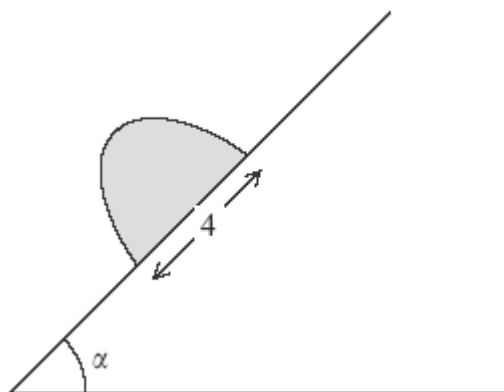
- (a) A uniform lamina is bounded by the curve $y = 4 - x^2$ and the x -axis, as shown in the diagram.



Given that the area of the lamina is $\frac{32}{3}$ square units, find the y -coordinate of the centre of mass of the lamina.

(5)

- (b) The cross-section of a uniform prism is the same shape as the lamina in part (a). The prism is placed on a plane inclined at an angle α to the horizontal with the rectangular base of the prism in contact with the inclined plane, as shown in the diagram.



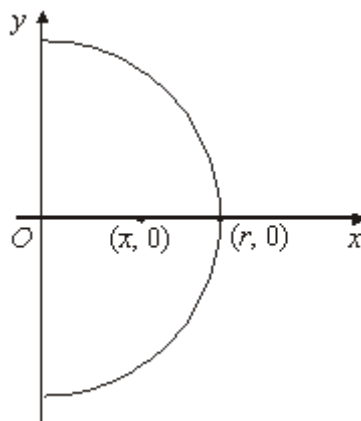
Given that the prism is just about to topple and that no slipping occurs, find the value of α , giving your answer to the nearest degree.

(4)

(Total 9 marks)

Q4.

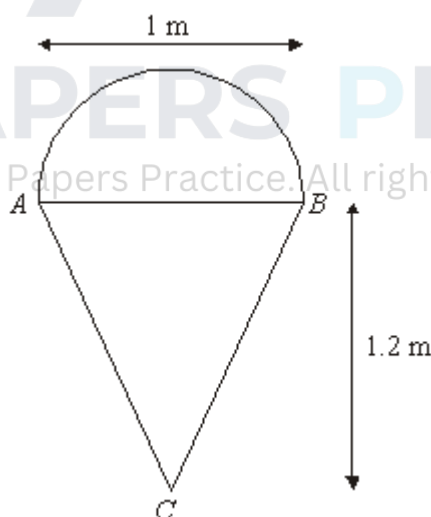
- (a) A uniform semicircular lamina of radius r has its centre at the origin and its axis of symmetry along Ox . The position of its centre of mass has coordinates $(\bar{x}, 0)$.



- (i) Show that $\frac{1}{2} \pi r^2 \bar{x} = \int_0^r 2x\sqrt{r^2 - x^2} dx$. (4)

- (ii) Hence prove that $\bar{x} = \frac{4r}{3\pi}$. (3)

- (b) The diagram below shows a uniform lamina, used as an advertising feature in a local supermarket to promote ice cream. It consists of a semicircle, of diameter $AB = 1$ m, and an isosceles triangle ABC , where C is at a distance 1.2 m from AB .



- (i) State the distance of the centre of mass of the **triangle** from C . (1)
- (ii) Show that the distance of the centre of mass of the **semicircle** from C is approximately 1.41 m. (1)
- (iii) Find the distance of the centre of mass of the complete lamina from C .

(4)

- (c) The lamina is freely suspended from A . Find the angle that AB makes with the vertical through A , giving your answer to the nearest degree.

(3)

(Total 16 marks)

Q5.

A uniform lamina is bounded by the x -axis, and the lines $x = a$ and $y = kx$, where k is a positive constant.

- (a) Find the area of the lamina in terms of a and k .

(2)

- (b) Use integration to show that the x -coordinate of the centre of mass of the lamina is $\frac{2a}{3}$.

(4)

- (c) Use integration to find the y -coordinate of the centre of mass of the lamina.

(4)

(Total 10 marks)

Q6.

A uniform triangular lamina has vertices at the points with coordinates $A(0, 0)$, $B(9, 0)$ and $C(9, 6)$.

- (a) Find the equation of the line that passes through the points A and C .

(2)

- (b) (i) Use integration to show that the x -coordinate of the centre of mass of the lamina is 6.

(4)

- (ii) Use integration to find the y -coordinate of the centre of mass of the lamina.

(4)

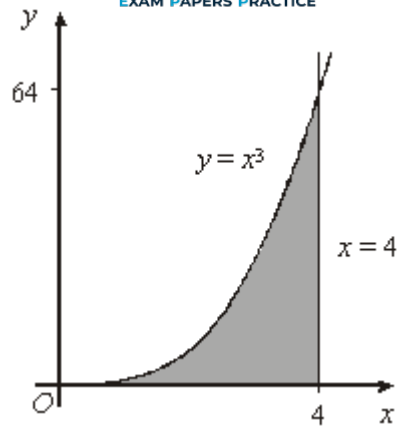
- (c) The lamina is suspended in equilibrium from the corner A . Find the angle between the side AB and the vertical.

(3)

(Total 13 marks)

Q7.

A uniform lamina is bounded by the curve $y = x^3$, the line $x = 4$ and the x -axis.



- (a) Find the area of the lamina. (2)
- (b) Use integration to show that the x -coordinate of the centre of mass of the lamina is 3.2. (4)
- (c) Find the y -coordinate of the centre of mass of the lamina. (4)
- (d) The lamina is suspended in equilibrium from its right-angled corner. Find the angle between the longer of the two straight sides of the lamina and the vertical. (4)

(Total 14 marks)